

Mitigating Web Application Vulnerabilities

Internet and Web Application Security 3e - Lab 4

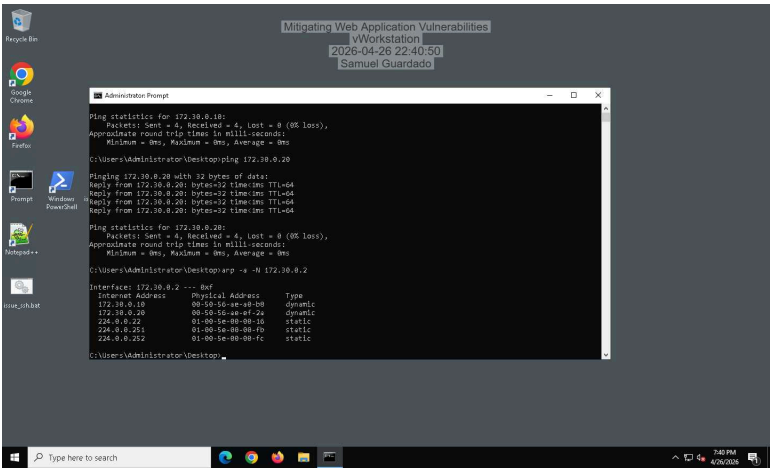
Student: Samuel Guardado
Email: sguar5@brockport.edu
Report Generated: Thursday, July 30, 2026 at 5:19 PM

Time on Task: 1 hour, 21 minutes
Progress: 100%

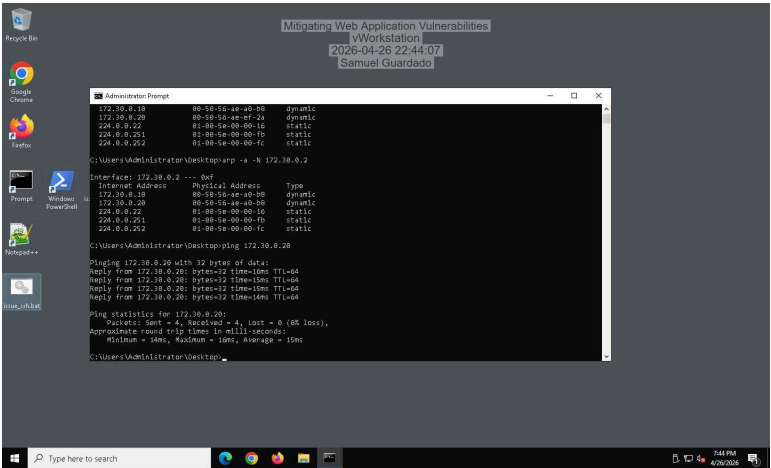
Section 1: Hands-On Demonstration

Part 1: Mitigate an HTTP MITM Attack by Using HTTPS

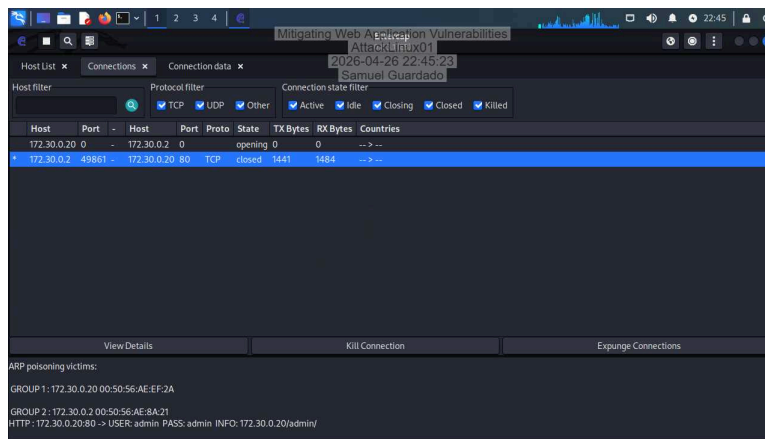
11. Make a screen capture showing the physical addresses for 172.30.0.10 and 172.30.0.20.



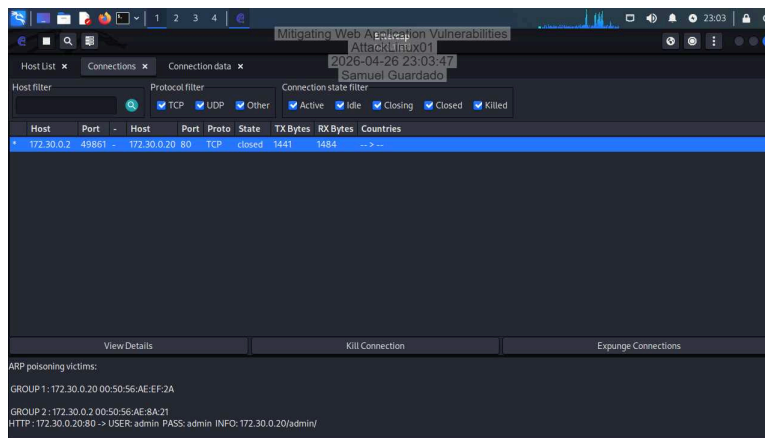
27. Make a screen capture showing the ARP information and the successful ping.



34. Make a screen capture showing the username and password in the Ettercap log.

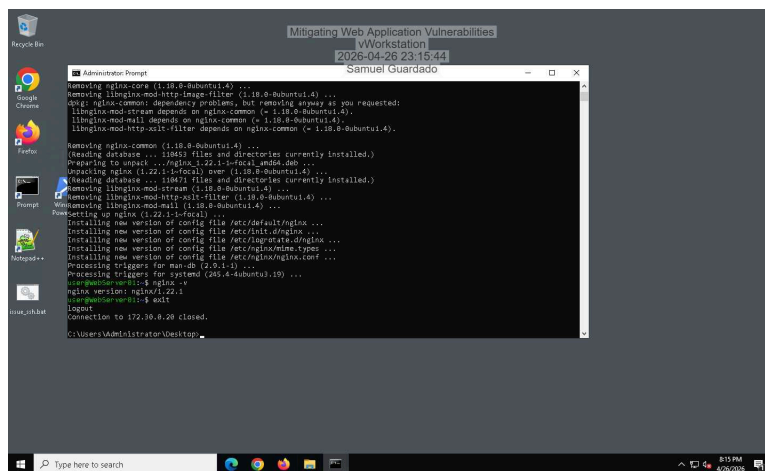


55. Make a screen capture showing the HTTPS connection in Ettercap.



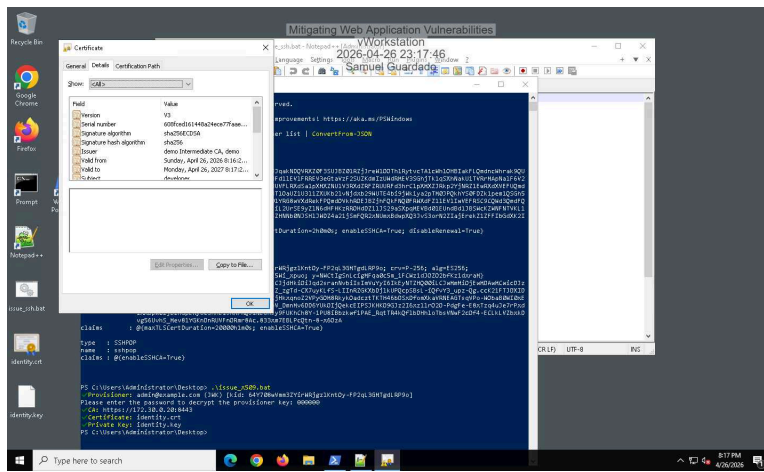
Part 2: Mitigate Potential Future Attacks by Updating the Web Server

10. Make a screen capture showing the new nginx version.

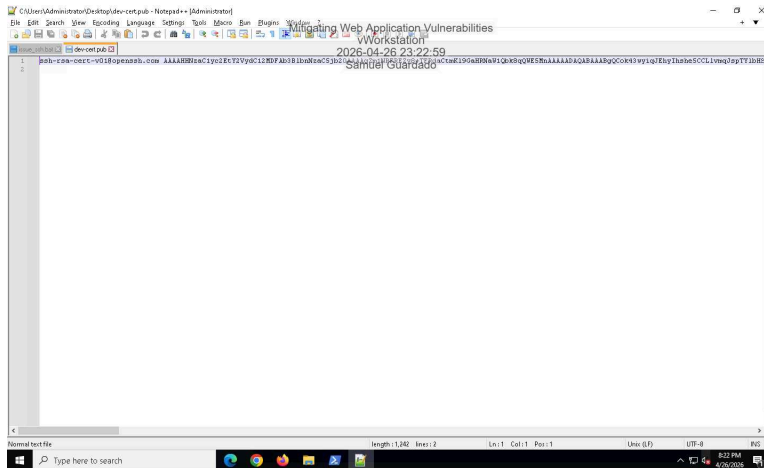


Part 3: Mitigate Unauthorized Changes Using Access Control

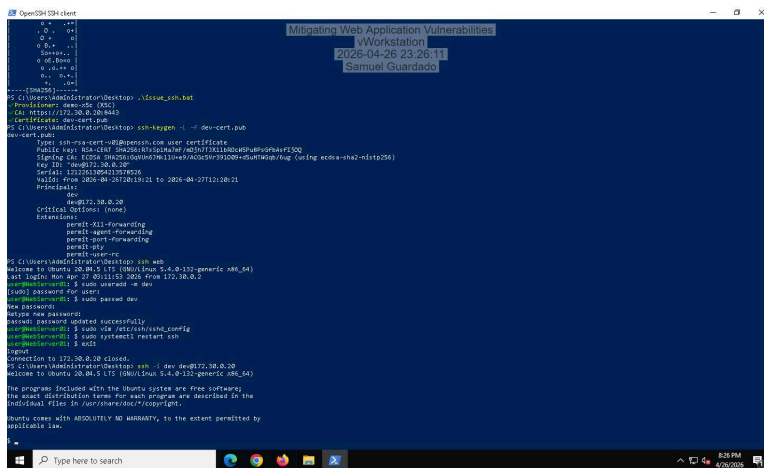
8. Make a screen capture showing the details of the certificate, including the Subject. You may have to enlarge the window to do this.



14. Make a screen capture showing the details of the SSH certificate.



27. Make a screen capture showing the successful login command and prompt on WebServer01.



Section 2: Challenge and Analysis

Part 1: Disable Password Access for Dev

1. Document the changes required in the sshd_config file.

I added: Match User Dev PasswordAuthentication no

2. Make a screen capture showing a rejected login when no key or certificate is used.

```
user@WebServer01:~$ cat /etc/ssh/sshd_config
# Configuration file for the sshd daemon.
# See sshd(8) for details.

# Port 22
Port 22

# Address family
AddressFamily inet

# Listen on all interfaces (unless ListenAddress is set).
Listen 0.0.0.0

# AllowTcpForwarding yes
# X11Forwarding yes
# PermitTTY yes
# PasswordAuthentication yes
# PubkeyAuthentication yes
# Match User Dev PasswordAuthentication no
```

Part 2: Disable User-Supplied Authorized_keys for Dev

1. Document the changes required in the sshd_config file.

I added: the three processes that connected to 205.134.253.10:4444.

2. Make a screen capture showing a rejected login when no certificate is used.

```
user@WebServer01:~$ cat /etc/ssh/sshd_config
# Configuration file for the sshd daemon.
# See sshd(8) for details.

# Port 22
Port 22

# Address family
AddressFamily inet

# Listen on all interfaces (unless ListenAddress is set).
Listen 0.0.0.0

# AllowTcpForwarding yes
# X11Forwarding yes
# PermitTTY yes
# PasswordAuthentication yes
# PubkeyAuthentication yes
# Match User Dev PasswordAuthentication no

# AuthorizedKeysFile .ssh/authorized_keys
# Match User Dev AuthorizedKeysFile none
```

```
PS C:\Users\Administrator\Desktop> ssh web
Welcome to Ubuntu 20.04.5 LTS (GNU/Linux 5.4.0-132-generic x86_64)
Last login: Mon Apr 27 03:11:53 2026 from 172.30.0.2
user@WebServer01:~$ sudo useradd -m dev
[sudo] password for user:
user@WebServer01:~$ sudo passwd dev
New password:
Retype new password:
passwd: password updated successfully
user@WebServer01:~$ sudo vim /etc/ssh/sshd_config
user@WebServer01:~$ sudo systemctl restart ssh
user@WebServer01:~$ exit
logout
Connection to 172.30.0.20 closed.
PS C:\Users\Administrator\Desktop> ssh -i dev dev@172.30.0.20
Welcome to Ubuntu 20.04.5 LTS (GNU/Linux 5.4.0-132-generic x86_64)

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

$ exit
Connection to 172.30.0.20 closed.
PS C:\Users\Administrator\Desktop> ssh web
Welcome to Ubuntu 20.04.5 LTS (GNU/Linux 5.4.0-132-generic x86_64)
Last login: Mon Apr 27 03:23:14 2026 from 172.30.0.2
user@WebServer01:~$ sudo vim /etc/ssh/sshd_config
[sudo] password for user:
user@WebServer01:~$ sudo systemctl restart ssh
user@WebServer01:~$ ssh deb@172.30.0.20
The authenticity of host '172.30.0.20 (172.30.0.20)' can't be established.
ECDSA key fingerprint is SHA256:PDc9/WH/vi821W1GuE2pr227h1Fv2HYLYvWZM20CqM4.
Are you sure you want to continue connecting (yes/no/[fingerprint])? y
Please type 'yes', 'no' or the fingerprint: yes
Warning: Permanently added '172.30.0.20' (ECDSA) to the list of known hosts.
deb@172.30.0.20's password:
Connection closed by 172.30.0.20 port 22
user@WebServer01:~$ sudo vim /etc/ssh/sshd_config
user@WebServer01:~$ ssh -i dev dev@172.30.0.20
Warning: Identity file dev not accessible: No such file or directory.
dev@172.30.0.20: Permission denied (publickey).
user@WebServer01:~$
```